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MACHINE LEARNING · COMPANION READER

Session 15: Support Vector Machines (SVM)

Maximum Margin, Primal/Dual Formulations, KKT Conditions, Soft Margin, and Kernels

Read alongside the lecture slides · Every slide example worked out in full

1 Introduction to Linear Classifiers & Maximum Margin

Finding the optimal separating hyperplane that maximizes the geometric distance to the closest training points.

A linear classifier separates data points belonging to two classes ($y_i \in \{-1, +1\}$) using a decision boundary (hyperplane):

$$f(x) = \text{sign}(w^T x + b) \quad (1)$$

where w is the weight vector normal to the hyperplane and b is the bias term.

While many hyperplanes can separate linearly separable data, the **Support Vector Machine (SVM)** selects the unique hyperplane that maximizes the **margin** (the distance between the decision boundary and the closest data points from either class).

The intuition

Think of building a highway between two towns (classes). SVM constructs the widest possible highway so that vehicles stay as far away from town borders as possible.

2 Primal SVM Optimization Problem

Formulating maximum margin classification as a constrained quadratic optimization problem.

The geometric margin distance from a point x_i to hyperplane $w^T x + b = 0$ is $\frac{y_i(w^T x_i + b)}{\|w\|}$. Scaling w and b such that $y_i(w^T x_i + b) = 1$ for the closest points gives a total margin width of $\frac{2}{\|w\|}$.

Maximizing $\frac{2}{\|w\|}$ is equivalent to minimizing $\frac{1}{2}\|w\|^2$.

2.1 Hard Margin Primal Formulation

$$\min_{w, b} \frac{1}{2}\|w\|^2 \quad \text{subject to } y_i(w^T x_i + b) \geq 1, \quad \forall i = 1, \dots, N \quad (2)$$

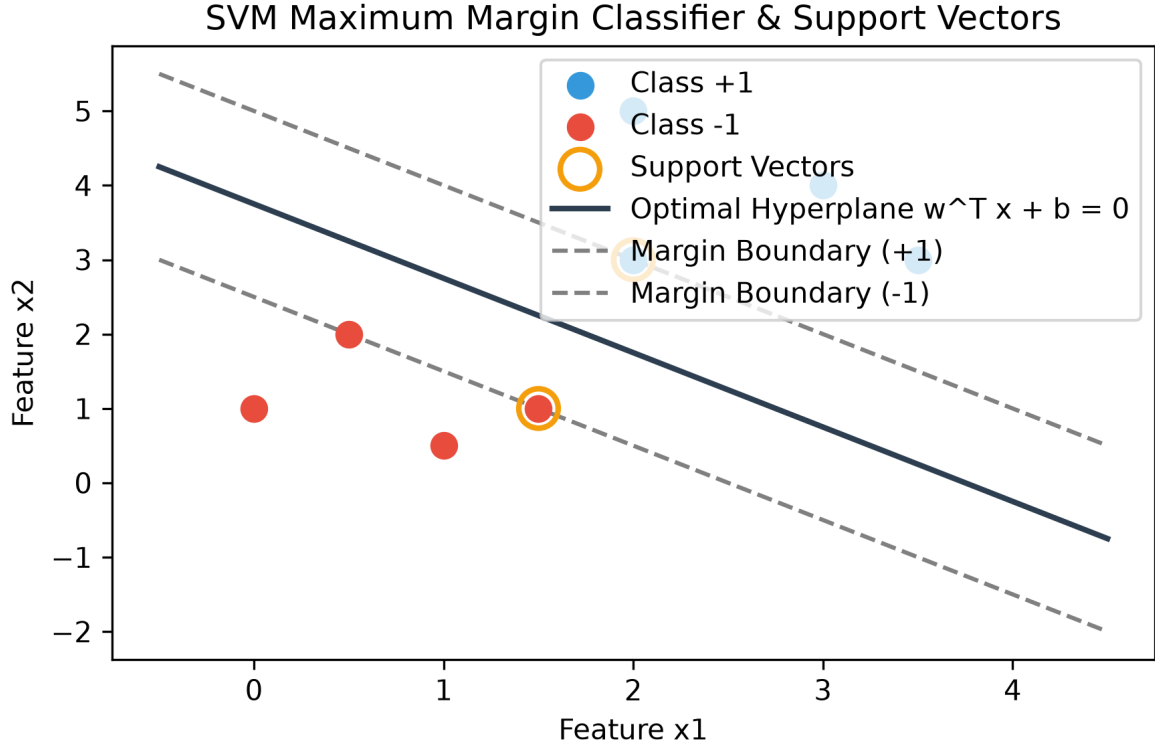


Figure 1: SVM maximum margin boundary and highlighted support vectors controlling hyperplane orientation.

3 Lagrangian Duality and KKT Conditions

Converting constrained primal optimization into an unconstrained dual problem via Lagrange multipliers.

Using non-negative Lagrange multipliers $\alpha_i \geq 0$, the Lagrangian function is:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^N \alpha_i [y_i(w^T x_i + b) - 1] \quad (3)$$

Setting derivatives with respect to primal variables w and b to zero yields:

$$\frac{\partial L}{\partial w} = 0 \implies w = \sum_{i=1}^N \alpha_i y_i x_i \quad (4)$$

$$\frac{\partial L}{\partial b} = 0 \implies \sum_{i=1}^N \alpha_i y_i = 0 \quad (5)$$

3.1 Karush-Kuhn-Tucker (KKT) Conditions

The optimal solution must satisfy complementary slackness:

$$\alpha_i [y_i(w^T x_i + b) - 1] = 0, \quad \forall i \quad (6)$$

This implies that $\alpha_i > 0$ ONLY for points on the margin boundary ($y_i(w^T x_i + b) = 1$). These critical points are called **Support Vectors**.

3.2 Dual Optimization Problem

Substituting $w = \sum \alpha_i y_i x_i$ back into the Lagrangian produces the Wolfe Dual:

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j (x_i^T x_j) \quad \text{s.t.} \quad \alpha_i \geq 0, \sum_{i=1}^N \alpha_i y_i = 0 \quad (7)$$

4 Soft Margin SVM and Slack Variables

Allowing controlled misclassifications in noisy data using slack variables ξ_i and penalty parameter C .

Real-world data is rarely perfectly separable. We introduce non-negative slack variables $\xi_i \geq 0$ allowing samples to fall inside the margin or be misclassified.

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (8)$$

The parameter $C > 0$ controls the trade-off: large C penalizes misclassifications heavily (narrower margin), while small C tolerates errors for a wider margin.

5 The Kernel Trick and Non-linear SVM

Implicitly mapping low-dimensional input vectors into high-dimensional feature space using kernel functions.

When data is non-linearly separable, we map $x \rightarrow \Phi(x)$ into a higher-dimensional space where linear separation is possible.

By Mercer's Theorem, instead of explicitly calculating $\Phi(x_i)^T \Phi(x_j)$, we replace dot products with a kernel function $K(x_i, x_j) = \Phi(x_i)^T \Phi(x_j)$:

- **Polynomial Kernel:** $K(x, z) = (x^T z + c)^d$
- **Gaussian RBF Kernel:** $K(x, z) = \exp(-\gamma \|x - z\|^2)$

Worked example: 1D SVM Dual Solution

Let us solve a 1D SVM numerical problem step-by-step:

Given points $x_1 = 1(y_1 = -1)$ and $x_2 = 3(y_2 = +1)$.

Step 1. Dual Equation: $\max \alpha_1 + \alpha_2 - \frac{1}{2} [\alpha_1^2(1)(1) + 2\alpha_1\alpha_2(-1)(+1)(3) + \alpha_2^2(9)]$ Constraint: $-\alpha_1 + \alpha_2 = 0 \implies \alpha_1 = \alpha_2 = \alpha$.

Step 2. Optimize Alpha: Substitute α : $f(\alpha) = 2\alpha - \frac{1}{2} [\alpha^2 - 6\alpha^2 + 9\alpha^2] = 2\alpha - 2\alpha^2$. Derivative: $\frac{df}{d\alpha} = 2 - 4\alpha = 0 \implies \alpha = 0.5$.

Step 3. Compute Weight w and Bias b : $w = \alpha_1 y_1 x_1 + \alpha_2 y_2 x_2 = 0.5(-1)(1) + 0.5(+1)(3) = 1.0$. Using support vector x_2 : $y_2(w x_2 + b) = 1 \implies 1(1(3) + b) = 1 \implies b = -2$. Final Hyperplane: $f(x) = x - 2 = 0 \implies x = 2$.

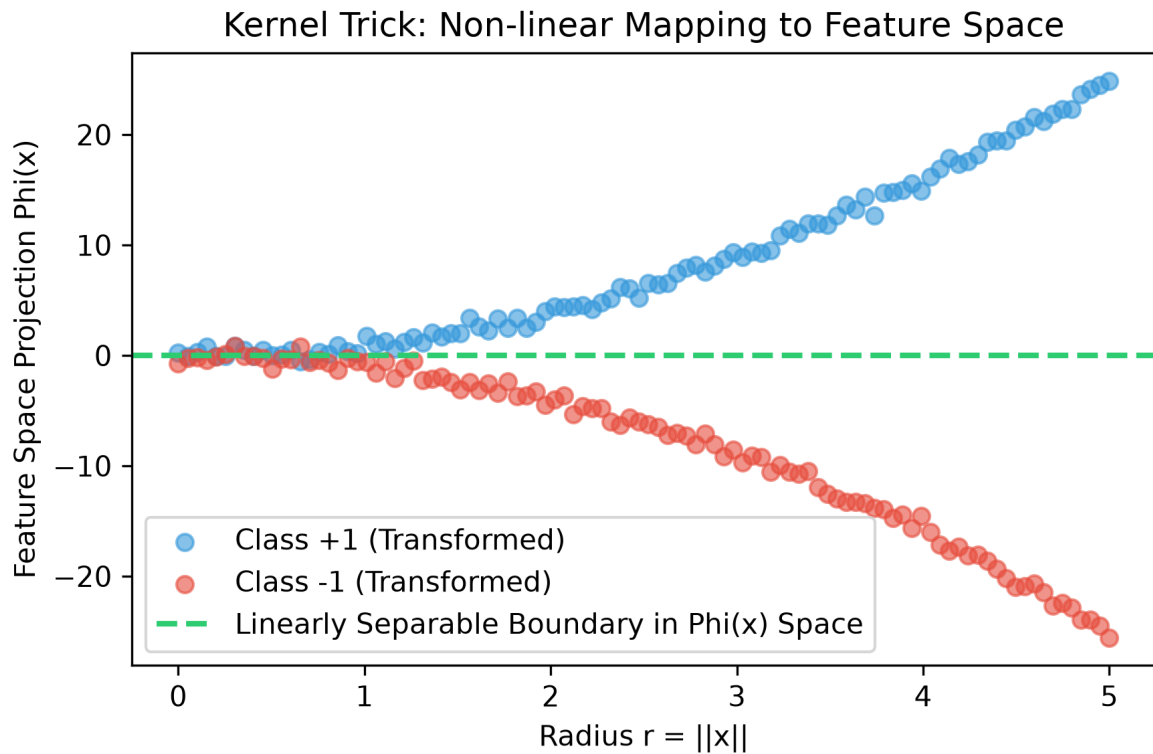


Figure 2: Kernel transformation maps non-linear 2D data into a linearly separable 3D space.

6 Summary & Self-Test

✓ Key takeaway

SVM maximizes margin $2/\|w\|$, expressed in dual form via Lagrange multipliers. Non-linear boundaries are efficiently created via the Kernel trick without explicitly computing high-dimensional features.

6.1 Self-Test Quiz

Q1: What is a support vector?

Answer: A data point lying exactly on the margin boundary ($\alpha_i > 0$) that defines the hyperplane.